

Conditional Expectation

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Overview

Great Expectations (18.4)

 Alternate Definition of Expectation (18.4.4)

 Conditional Expectation (18.4.5)

 Mean Time to Failure (18.4.6)

 Expected Returns in Gambling Games (18.4.7)

Shortcut formula for expectation

Instead of summing over all outcomes, sometimes easier to group outcomes by the blocks defined by the random variable.

$$\mathbb{E}[R] = \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x].$$

Proof:

$$\begin{aligned}
 \mathbb{E}[R] &= \sum_{\omega \in \mathcal{S}} R(\omega) \Pr[\omega] && \text{defn expt} \\
 &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} R(\omega) \Pr[\omega] && \text{range is partition} \\
 &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} x \Pr[\omega] && \text{defn } R, \omega \\
 &= \sum_{x \in \text{range}(R)} x \sum_{\omega \in [R=x]} \Pr[\omega] && \text{factor out } x \\
 &= \sum_{x \in \text{range}(R)} x \Pr[R = x]. && \text{defn event prob}
 \end{aligned}$$

Median

Not the same as mean or expected value. It's the probability-weighted middle of the set of values.

Definition: The *median* of a random variable R is the value $x \in \text{range}(R)$ such that

$$\Pr[R \leq x] \leq \frac{1}{2}$$

and

$$\Pr[R \geq x] < \frac{1}{2}.$$

Book: “We won’t devote much attention to the median. The expected value is more useful and has much more interesting properties.”

Me: Pthh.

Definition of conditional expectation

Definition: The conditional expectation of a random variable R given event A is:

$$\mathbb{E}[R|A] := \sum_{r \in \text{range}(R)} r \cdot \Pr[R = r|A].$$

Example: R is the deck of cards example from before: value of card, 1 for ace, 10 for face.

$$\begin{aligned}\mathbb{E}[R|R < 10] &= \sum_{i=1}^9 i \Pr[R = i|R < 10] \\&= \sum_{i=1}^9 i \frac{\Pr[R=i \text{ AND } R < 10]}{\Pr[R < 10]} \\&= \sum_{i=1}^9 i \frac{1/13}{9/13} \\&= 1/9 \sum_{i=1}^9 i \\&= 45/9 = 5.\end{aligned}$$

The height of usefulness

How tall are elephants on average? Don't know. Web search didn't help. Conditional expectations can!

$$\begin{aligned}
 \mathbb{E}[H] &= \mathbb{E}[H|Af] \cdot \Pr[Af] + \mathbb{E}[H|As] \cdot \Pr[As] && \text{split} \\
 &= 10 \cdot 0.9 + \mathbb{E}[H|As] \cdot 0.1 && \text{Google} \\
 &= 10 \cdot 0.9 + (\mathbb{E}[H|As \text{ AND } m] \Pr[m|As] \\
 &\quad + \mathbb{E}[H|As \text{ AND } f] \cdot \Pr[f|As]) \cdot 0.1 && \text{split} \\
 &= 10 \cdot 0.9 + (9 \cdot 1/2 + 7 \cdot 1/2) \cdot 0.1 && \text{Google} \\
 &= 9.8 && \text{calculator}
 \end{aligned}$$

Memoryless crash time

A computer program crashes at the end of each hour of use with probability p , if it has not crashed already. Let C be the random variable for the time until a crash occurs. What is $\mathbb{E}[C]$?

Let A be the event that the program crashes after the first hour and $\bar{A}$ be the complementary event. Mean time to failure is

$$\mathbb{E}[C] = \mathbb{E}[C|A] \cdot \Pr[A] + \mathbb{E}[C|\bar{A}] \cdot \Pr[\bar{A}]$$

$\mathbb{E}[C|A] = 1$ because A is the event that the program crashes after one hour.

$\mathbb{E}[C|\bar{A}] = 1 + \mathbb{E}[C]$ because the program runs for an hour, then we're back in the original situation.

$$\mathbb{E}[C] = 1 \cdot p + (1 + \mathbb{E}[C])(1 - p)$$

$$\mathbb{E}[C] = p + 1 + \mathbb{E}[C] - p - p\mathbb{E}[C]$$

$$p\mathbb{E}[C] = 1$$

$$\mathbb{E}[C] = 1/p$$

Mean babies until boy

If a couple decides to keep having babies until one is a boy, assuming there are just boys and girls and they are equally likely, how many kids will they make?

failure	boy
hour	baby
p	$1/2$
$1/p$	2

- ▶ Coin flips until heads? 2
- ▶ Cards until face card? $13/3 = 4.3$ ish
- ▶ Kobe (0.85) free throws until basket? 1.18

Expected winnings

D dollars won, w event that you win, $\bar{w}$ event that you lose.

$$\mathbb{E}[D] = \mathbb{E}[D|w] \Pr[w] + \mathbb{E}[D|\bar{w}](1 - \Pr[w])$$

Game is *fair* if $\mathbb{E}[D] = 0$.

Example: $\mathbb{E}[D|w] = -\mathbb{E}[D|\bar{w}]$, $\Pr[w] = 1/2$.

Example: SAT points (before March 2016). $\mathbb{E}[D|w] = 1$,
 $\mathbb{E}[D|\bar{w}] = -1/4$, $\Pr[w] = 1/5$.

$$1 \cdot .2 + (-1/4) \cdot .8 = 0.$$

Bar bet

Three people. Each player puts in \$2. Each player predicts the outcome of a coin flip. Correct guessers split the pot. If all wrong, money returned.

$$\begin{aligned}\mathbb{E}[D] &= \mathbb{E}[D|\text{win alone}] \Pr[\text{win alone}] \\ &\quad + \mathbb{E}[D|\text{win with one}] \Pr[\text{win with one}] \\ &\quad + \mathbb{E}[D|\text{win with both}] \Pr[\text{win with both}] \\ &\quad + \mathbb{E}[D|\text{lose}] \Pr[\text{lose}] \\ &\quad + \mathbb{E}[D|\text{all lose}] \Pr[\text{all lose}]\end{aligned}$$

$$\begin{aligned}\mathbb{E}[D] &= 4 \cdot \Pr[\text{win alone}] \\ &\quad + 1 \cdot \Pr[\text{win with one}] \\ &\quad + 0 \cdot \Pr[\text{win with both}] \\ &\quad - 2 \cdot \Pr[\text{lose}] \\ &\quad + 0 \cdot \Pr[\text{all lose}]\end{aligned}$$

Random guesses

$$\begin{aligned}\mathbb{E}[D] &= 4 \cdot \Pr[\text{win alone}] \\ &\quad + 1 \cdot \Pr[\text{win with one}] \\ &\quad + 0 \cdot \Pr[\text{win with both}] \\ &\quad - 2 \cdot \Pr[\text{lose}] \\ &\quad + 0 \cdot \Pr[\text{all lose}]\end{aligned}$$

Everyone guesses randomly.

- ▶ $\Pr[\text{win alone}] = 1/8$
- ▶ $\Pr[\text{win with one}] = 1/4$
- ▶ $\Pr[\text{win with both}] = 1/8$
- ▶ $\Pr[\text{lose}] = 3/8$
- ▶ $\Pr[\text{all lose}] = 1/8$

$$4 \cdot 1/8 + 1 \cdot 1/4 + 0 \cdot 1/8 - 2 \cdot 3/8 + 0 \cdot 1/8 = 0. \text{ Fair!}$$

Collusion

$$\begin{aligned}\mathbb{E}[D] &= 4 \cdot \Pr[\text{win alone}] \\ &\quad + 1 \cdot \Pr[\text{win with one}] \\ &\quad + 0 \cdot \Pr[\text{win with both}] \\ &\quad - 2 \cdot \Pr[\text{lose}] \\ &\quad + 0 \cdot \Pr[\text{all lose}]\end{aligned}$$

The other two collude to make opposite guesses.

- ▶ $\Pr[\text{win alone}] = 0$
- ▶ $\Pr[\text{win with one}] = 1/2$
- ▶ $\Pr[\text{win with both}] = 0$
- ▶ $\Pr[\text{lose}] = 1/2$
- ▶ $\Pr[\text{all lose}] = 0$

$$4 \cdot 0 + 1 \cdot 1/2 + 0 \cdot 0 - 2 \cdot 1/2 + 0 \cdot 0 = -0.5. \text{ I lose!}$$